

Analyzing Students' Use of Generative Artificial Intelligence in Collaborative Problem Solving

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Abstract

This study analyzed how students used generative artificial intelligence (GAI) tools in collaborative problem solving. We examined students' GAI usage using conversational data and screen recordings from 12 groups, each comprising three university students. A coding framework was developed through hybrid thematic analysis to capture students' GAI use behaviors at both individual and group levels. Four types of behaviors were identified: prompting, information sharing, information use, and evaluation. Based on this framework, we conducted a correlation analysis to examine how these behaviors were related to students' knowledge construction. The results showed that GAI use behaviors involving sharing or comparing responses were correlated with lower levels of knowledge construction, whereas behaviors involving co-editing and evaluating were correlated with higher levels. This study contributes a new framework for analyzing individual and collective use of GAI in collaborative learning and offers pedagogical insights that guide students' effective use of GAI tools in teaching practice.

CCS Concepts

• Applied computing → Collaborative learning.

Keywords

ChatGPT; Collaborative problem solving; Generative artificial intelligence; Interaction Analysis Model; Knowledge Construction

ACM Reference Format:

Xiuyu Chen and Shihui Feng. 2025. Analyzing Students' Use of Generative Artificial Intelligence in Collaborative Problem Solving. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3715070.3749257>

1 Introduction

The term generative artificial intelligence (GAI) refers to computational techniques that are capable of generating new meaningful content such as text, images, or audio from training data (e.g., ChatGPT) [6]. GAI tools are increasingly applied in educational settings, where they can support student learning through functions such as

answering questions, summarizing information, helping with drafting, and providing feedback [13]. The personalized nature of GAI, along with its ability to adapt learning experiences to the unique needs of individual students, could be integrated into collaborative learning environments where students learn at their own pace, in their own style, and come together to share ideas and perspectives [19]. GAI can serve as a learning assistant to support knowledge construction by providing personalized feedback, inspiring ideas, and offering on-demand access to educational resources [15]. In addition, GAI tools have the potential to facilitate collaborative learning by generating diverse scenarios in which students can work together to solve problems and achieve shared goals [16]. Understanding how students use GAI tools is especially important in collaborative learning contexts, where such technology not only supports individual use but also influences group interactions and knowledge construction. Collaborative problem solving (CPS) involves addressing a problem collectively through active collaboration and idea exchange, and has been recognized as a promising task that requires a range of social and cognitive skills [9]. Since CPS relies heavily on students' construction of new knowledge, it is essential to examine this process by analyzing students' dialogue and the tools that mediate their learning [10]. In this study, we developed a coding framework to model students' GAI use behaviors at both the individual and group levels during CPS tasks. We also examined how these behaviors relate to an existing framework of knowledge construction.

This study is guided by the following research questions: (1) What types of behaviors were identified in students' use of GAI tools during CPS? (2) How did students' GAI use behaviors relate to their knowledge construction during CPS?

2 Related Work

Knowledge construction refers to the processes through which students solve problems and develop understanding of concepts, phenomena, and situations, involving qualitative changes in the complexity of their thinking and conceptualization of subject matter [18]. To examine how our proposed coding framework aligns with an established model of knowledge construction, we analyzed the relationships between our GAI use behaviors and the Interaction Analysis Model [8]. The Interaction Analysis Model includes five phases of knowledge construction: (1) Sharing and comparing information: presenting new information to group members, indicating agreement, and expressing an observation or viewpoint. (2) Discovery and exploration of dissonance: identifying areas of disagreement, and asking and answering questions to clarify those disagreements. (3) Negotiation of meaning: clarifying or negotiating the meaning of terms, and suggesting and negotiating new

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CSCW Companion '25, Bergen, Norway

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ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3749257>

statements that reflect compromise and co-construction. (4) Testing and modification of synthesis: testing proposed new knowledge against existing cognitive schemas, personal experience, or other sources. (5) Application of meaning: summarizing agreements and applying new knowledge.

Previous studies have found that GAI tools offer learning opportunities by enabling discussions, consultations, and instant access to information that support knowledge construction [17, 19]. However, relying solely on input prompts may limit students' control over the knowledge construction process [14]. While prior studies have highlighted the potential role of GAI in students' knowledge construction, the relationship between students' GAI usage and their knowledge construction processes remains unclear.

3 Methodology

3.1 Learning Setting and Participants

This study involved 36 university students who participated in a CPS activity conducted in a laboratory setting. All participants were recruited from the same university and were divided into 12 groups, each comprising three members. Five groups consisted of master's students, while the remaining seven consisted of undergraduate students. Among the participants, 24 were female and 12 were male. Eleven students were from Science, Technology, Engineering, and Mathematics (STEM) majors, and the remaining 25 were from non-STEM majors.

Each CPS task lasted approximately 30 minutes. Each group was assigned an educational framework to study and completed both a set of multiple-choice questions and a group essay related to the assigned framework. Of the 12 groups, six focused on Bloom's Taxonomy, while the remaining six examined the Interactive, Constructive, Active, and Passive (ICAP) framework in education. Bloom's Taxonomy is designed to guide lesson planning by helping teachers set objectives that lead students from lower-order to higher-order thinking [7]. The ICAP framework is comparable to Bloom's Taxonomy [4], and both frameworks describe observable verbs that reflect student behaviors at different levels of learning objectives. These frameworks provided participants with a useful structure for studying task-related concepts during the CPS tasks. The multiple-choice questions required students to select verbs corresponding to each level of their assigned framework. For the group essay, students were asked to discuss how teachers can apply the assigned framework to support students in transitioning from low-level to high-level engagement. Two GAI platforms were made available during the CPS activity: an AI chatbot aggregator and ChatGPT. Students had access to both ChatGPT 3.5 and ChatGPT 4 through these platforms. Students were encouraged to use the GAI tools during the activity but were also allowed to consult alternative information sources if they preferred.

3.2 Data Collection

This study collected multimodal process data during the CPS activities to analyze participants' GAI usage. The dataset included three components: (1) Screen recordings, which captured participants' interactions with GAI tools and other information sources, provided detailed insights into how students engaged with these

tools. (2) Conversational data from video recordings, which documented group discussions, was used to code students' knowledge construction as well as how they utilized and integrated GAI tools at the group level. (3) A post-test questionnaire, where students rated the perceived difficulty of the tasks, was used to control for differences in task difficulty across conditions. These differences could potentially influence how students engaged with GAI tools.

3.3 Data Analysis

To address RQ1, we conducted a hybrid thematic analysis [2] to synthesize students' GAI use behaviors at the individual and group levels during CPS processes, based on screen recordings and video recordings of group discussions. This method integrated both deductive (theory-informed) and inductive (data-informed) coding approaches. Deductive coding was guided by the theoretical framework of collaborative information behavior [12], ensuring that initial codes were grounded in established concepts, while inductive coding allowed additional codes to emerge directly from the data. We refined the procedure of the hybrid thematic analysis [2] to adapt it to our dataset and analyzed the data as follows: (1) Familiarization with the multimodal dataset: We reviewed the multimodal dataset to become familiar with how students engaged with GAI tools during the CPS tasks. We ensured consistency across modalities by aligning screen and video recording data using shared timestamps and observable events. (2) Generating initial codes: Initial coding was both theory- and data-informed, following a cyclical, iterative process. Our coding scheme was grounded in the collaborative information behavior model [12], which includes three key phases: problem formulation, collaborative information seeking, and information use, each comprising several constitutive activities. Drawing on this model, we identified four observable behavior types: prompting, information sharing, information use, and evaluation. Prompting was considered part of information seeking, where students used GAI tools as information sources. (3) Developing behavioral indicators: This step involved organizing and combining codes to form broader categories. In this step, we developed two subcategories: individual level and group level, under each of the four categories derived from the collaborative information behavior framework. We found that students exhibited different types of behaviors when using GAI tools across the two data sources: screen recordings captured individual use, whereas video recordings reflected collective use during group discussions. At this stage, the codes were initially organized into the corresponding categories. (4) Reviewing the candidate indicators: Some candidate indicators were merged due to conceptual overlap, while others were refined and separated into distinct indicators. Each indicator was required to be conceptually clear and distinguishable from the others. (5) Defining and naming categories and indicators: This step involved determining the essence of each indicator and organizing them into a coherent framework. We also defined and refined the indicators to ensure they were conceptually clear and suitable for analysis. Two coders independently coded the data, and intercoder reliability was assessed using Cohen's kappa [5] to evaluate their level of agreement. With a set of fully established indicators, we provided descriptions and selected examples that captured the meaning of the indicators.

To address RQ2, we constructed time series for each GAI usage and knowledge construction indicator to examine the synchronization of patterns between GAI use behaviors and knowledge construction over time. To explore their relationships, we first generated time series for each indicator. The data were segmented into fixed-length 30-second time windows, which were found to be appropriate for capturing students' GAI use behaviors. This segmentation was applied consistently across all groups. For GAI use behaviors, we used binary coding (1 = presence, 0 = absence) to indicate whether a specific behavior occurred within a given time window. For knowledge construction indicators, we calculated the frequency of discourse units corresponding to each indicator within the 30-second time windows. Correlation analysis was then conducted to examine the relationships between the constructed time series [3]. Specifically, we computed Pearson correlation coefficients to examine the relationships between GAI use behaviors and knowledge construction indicators. When multiple statistical tests were performed, the probability of obtaining false positives (Type I errors) increased, thereby inflating the overall error rate and requiring correction to maintain a valid significance threshold [11]. To control for multiple comparisons, we applied a false discovery rate (FDR) correction to the p-values [1]. Additionally, a Mann-Whitney U test was conducted to compare the perceived difficulty of the two CPS tasks, showing no significant differences between them ($U = 124$, $p = .209 > .05$).

4 Results

4.1 Coding Framework for Students' GAI Use Behaviors

To analyze students' GAI usage at the individual and group levels during CPS, we adopted the hybrid thematic analysis approach described in Section 3.3. Individual-level use referred to observable behaviors where students interacted with GAI tools independently, while group-level use involved collaborative interactions with GAI tools among group members. We categorized GAI use behaviors into four types: prompting, information sharing, information use, and evaluation (see Table 1). Intercode reliability was assessed using Cohen's kappa and yielded a value of 0.913, indicating substantial agreement between the two coders.

The category "prompting" captured how students engaged with GAI tools to obtain information. Screen recordings showed that prompting was primarily observed at the individual level and included five behaviors: copying and pasting task questions to prompt GAI directly, requesting GAI to explain concepts, correcting GAI responses, refining previous responses, and supporting peripheral tasks (e.g., grammar checking). We also observed two types of collective prompting: (1) prompting GAI based on group discussions and (2) discussing strategies for prompting GAI. The first behavior involved leveraging insights from group conversations to craft relevant prompts for GAI, while the second focused on group members working together to develop strategies for generating effective prompts to support their tasks.

The category "information sharing" involved students exchanging knowledge about GAI within their groups. This category included (1) sharing experiences of GAI usage, such as challenges or

observations, and (2) sharing GAI responses, such as information or insights generated by GAI tools.

The category "information use" referred to how students applied GAI responses to solve CPS tasks. At the individual level, students copied and pasted GAI responses into shared documents. At the group level, we observed (1) discussing task-related content built on GAI responses, which involved interpreting GAI responses and using them as a foundation to develop solutions for CPS tasks, and (2) co-editing GAI responses, where group members collaborated to modify and refine GAI outputs to better align with the CPS task. While individual use typically involved direct copying, co-editing reflected active revision of GAI outputs to meet group goals.

The category "evaluation" referred to assessing the quality and relevance of GAI outputs. At the individual level, students sometimes used GAI tools to verify the accuracy of previously returned responses. At the group level, evaluation involved comparing multiple GAI responses to the same query, discussing their usefulness and relevance, and evaluating co-edits of the responses. These activities were important for validating GAI responses and ensuring high-quality collaborative outcomes.

4.2 Correlations Between Students' GAI Use Behaviors and Knowledge Construction

We applied Pearson correlation to analyze the relationships between students' GAI use behaviors and their knowledge construction, presenting both raw p-values and adjusted p-values for multiple testing correction (see Table 2). For clarity, only the adjusted p-values are presented in the results section. A significant correlation was found between "sharing GAI responses" and "sharing and comparing information" ($r = 0.249$, $p < .001$). These two indicators are conceptually similar, as both involve the sharing of information or knowledge within the group. This suggests that students who shared GAI responses were primarily engaged in the early phase of knowledge construction, which involves the initial exchange of ideas.

Two GAI use behaviors showed significant correlations with the phase of "discovery and exploration of dissonance". The behavior "discussing task-related content built on GAI responses" ($r = 0.180$, $p = .001$) indicates that when students discussed concepts or ideas relevant to the CPS tasks, they often encountered divergent understandings that required further clarification. In these cases, GAI tools were used to help clarify key terms and resolve conceptual inconsistencies. Similarly, "comparing GAI responses among group members" ($r = 0.170$, $p = .002$) reflects that students occasionally noticed discrepancies or inconsistencies between different GAI responses, which encouraged them to collaboratively resolve these differences and reach a more accurate shared understanding.

A moderate correlation was found between "discussing task-related content built on GAI responses" and "negotiation of meaning" ($r = 0.372$, $p < .001$). This suggests that GAI responses helped students clarify task concepts and refine the meanings of key terms. Through this process, students developed a shared understanding by elaborating on the GAI responses and aligning their perspectives.

Three GAI use behaviors were significantly correlated with the phase of "testing and modification of synthesis": "evaluating the

Table 1: Framework for Modeling Individual and Collective Use of GAI Tools in Collaborative Problem Solving

Category	Level	Indicators	Examples
Prompting	Individual	Copy and paste task questions to prompt GAI directly	A student copied and pasted the entire question or part of it to prompt ChatGPT.
		Request GAI to explain concepts	A student asked ChatGPT to explain the key concept of the CPS task.
		Correct GAI responses	A student clarified that the ICAP framework is an educational framework, not a business framework with the same abbreviation.
		Request GAI to refine previous responses	A student requested ChatGPT to provide a shorter response.
		Request GAI to support peripheral tasks	A student asked ChatGPT to count the words of a paragraph.
	Group	Prompt GAI based on group discussion	Groups discussed the definition of active mode and prompted ChatGPT about it together.
		Discuss strategies for prompting GAI	One student suggested that the group members ask ChatGPT for more detailed information and provide timely, constructive feedback.
Information Sharing	Group	Share experiences of GAI usage	Groups shared a connection problem encountered while using ChatGPT.
		Share GAI responses	Groups shared the answer to a multiple-choice question given by ChatGPT.
Information Use	Individual	Copy and paste GAI responses	A student copied and pasted ChatGPT's responses into the group document.
	Group	Co-edit GAI responses	Groups revised ChatGPT's responses together by deleting and adding new content.
		Discuss task-related content built on GAI responses	Groups discussed the interpretation of different engagement modes based on ChatGPT's responses.
Evaluation	Individual	Check the accuracy of GAI responses	A student asked ChatGPT to assess the accuracy of a previous response.
	Group	Compare GAI responses among group members	Groups collaboratively compared the responses by asking ChatGPT the same question and then discussing the variations within their groups.
		Evaluate GAI responses with group members	Groups discussed the usefulness of a ChatGPT response together.
		Evaluate the co-edits of GAI responses	Groups evaluated their co-edits of GAI responses and discussed ways to improve it.

co-edits of GAI responses" ($r = 0.358$, $p < .001$), "co-editing GAI responses" ($r = 0.170$, $p = .002$), and "evaluating GAI responses with group members" ($r = 0.158$, $p = .004$). These behaviors primarily occurred during the group essay writing stage, when students collaboratively developed written outputs for CPS tasks. The results suggest that students did not simply accept GAI outputs but engaged in iterative processes of editing and evaluation. This reflects a deeper level of knowledge construction, as students revised GAI outputs in alignment with group goals through critical and constructive dialogue.

Both "evaluating the co-edits of GAI responses" ($r = 0.189$, $p < .001$) and "co-editing GAI responses" ($r = 0.194$, $p < .001$) were positively correlated with the phase of "application of meaning". These behaviors mainly occurred during the group essay writing stage, where students collaboratively applied their constructed understanding to generate solutions for the CPS tasks. In this phase,

students used what they had learned in earlier discussions to develop new solutions for CPS tasks, often leading to the generation of new insights.

Although our correlation analysis revealed that group-level GAI use behaviors were significantly associated with various phases of knowledge construction, no significant correlations were found between individual-level GAI use behaviors and knowledge construction. This suggests that students' collective use of GAI tools during group discussions plays a more influential role in facilitating knowledge construction than individual use.

5 Conclusion

In this study, we developed a coding framework to model students' GAI usage in the context of CPS. The framework identified four types of GAI use behaviors: prompting, information sharing, information use, and evaluation at both the individual and group levels.

Table 2: Correlations Between Students' GAI Use Behaviors and Knowledge Construction

Knowledge construction	GAI use behaviors	<i>r</i>	raw <i>p</i>	adj. <i>p</i>
Sharing and comparing information	Share GAI responses	0.249	0.000	0.000
Discovery and exploration of dissonance	Discuss task-related content built on GAI responses	0.180	0.000	0.001
Discovery and exploration of dissonance	Compare GAI responses among group members	0.170	0.000	0.002
Negotiation of meaning	Discuss task-related content built on GAI responses	0.372	0.000	0.000
Testing and modification of synthesis	Evaluate the co-edits of GAI responses	0.358	0.000	0.000
Testing and modification of synthesis	Evaluate GAI responses with group members	0.158	0.001	0.004
Testing and modification of synthesis	Co-edit GAI responses	0.170	0.000	0.002
Application of meaning	Evaluate the co-edits of GAI responses	0.189	0.000	0.000
Application of meaning	Co-edit GAI responses	0.194	0.000	0.000

The collective use of GAI tools reshapes peer interactions in collaborative learning, as students build on the information generated by these tools to interpret problems and collaboratively develop solutions for CPS tasks. We also examined the relationships between the GAI use behaviors and the phases of knowledge construction defined in the Interaction Analysis Model, and identified several significant correlations. The results indicated that GAI use behaviors involving sharing or comparing responses were linked to lower levels of knowledge construction, whereas behaviors involving co-editing and evaluating responses were associated with higher levels. This study highlights the importance of social context in GAI usage, suggesting that students' knowledge construction occurs more through collaborative engagement with GAI tools than through individual use alone.

Theoretically, this framework offers a valuable tool for examining how students engage with GAI tools in collaborative learning contexts. Pedagogically, this study offers practical implications for designing scaffolding strategies that support effective use of GAI tools and promote knowledge construction. Future studies could apply this framework to analyze how students from different educational levels use GAI tools, helping to identify their behavior patterns and common strategies in collaborative learning. This framework also provides a structured approach to understanding students' GAI use behaviors, which enables future researchers to conduct automated coding and analyze large volumes of student interaction data on GAI usage.

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